

Projecting seasonal African elephant connectivity under future climate and development scenarios identifies protected areas at risk of isolation

Olivia Hemond^{1*}, Evan Patrick^{1*}, Dana Cohen-Kaplan^{1,2}, Vikesh Dheeriy¹, Sam Muir^{1,3}, Noah Moyer¹, Nickolas McManus⁴, Joana Krieger⁴, Ezequiel Fabiano⁵, Colgar Sikopo⁶, Ashley Larsen¹, Patrick Roehrdanz⁴

*Indicates equal contribution

Author affiliations

1 Bren School of Environmental Science & Management, University of California, Santa Barbara, 93106-5131, USA.

2 Anvil Capture Systems, Inc., Somerville, MA, USA

3 Blue Forest, Sacramento, CA, USA

4 The Moore Center for Science and Solutions, Conservation International, 2011 Crystal Dr., Arlington, VA, 22202, USA.

5 Department of Wildlife Management and Tourism Studies, University of Namibia, Katima Mulilo 1096, Namibia.

6 Republic of Namibia Ministry of Environment, Forestry and Tourism, Windhoek 13306, Namibia

Corresponding author:

Dr. Evan Patrick

epatrick@ucsb.edu

2400 Bren Hall

Santa Barbara, CA 93106-5131

Abstract

1. The combined impacts of climate change and human development threaten functional connectivity and the continued persistence of many species. Preserving long-term landscape connectivity requires an understanding not only of species' movements in the present day, including their variation across seasons, but also of potential future changes in movement patterns.
2. We provide a workflow for forecasting functional connectivity seasonally under coupled climate and land-use change scenarios between 2015 and 2050, and apply it to African savanna elephants (*Loxodonta africana*) in the Kavango-Zambezi Transfrontier Conservation Area (KAZA).
3. We find that future scenarios with rapid warming and high land conversion decrease elephant connectivity and increase isolation risk for core elephant habitats, with the greatest connectivity declines during the wet season. These impacts are most acute in central KAZA, impacting corridors between many of the largest protected areas in the region.
4. *Synthesis and applications.* While KAZA has demonstrated success at conserving species by promoting landscape connectivity, we find emerging threats to connectivity may jeopardize conservation gains. Our workflow provides a transferable approach to forecast connectivity that explicitly accounts for seasonality and future scenario uncertainty, enabling proactive corridor planning and protected-area network design under global change.

Keywords

African elephant, climate change, ephemeral water holes, global change, habitat connectivity

1. Introduction

The effectiveness of protected areas hinges not only on their extent, but whether they function as well-connected ecological networks. Despite this, connectivity remains a major shortfall: only a small fraction of terrestrial land is both protected and connected, leaving many protected areas functionally isolated within human-modified landscapes (Brennan et al., 2022; UNEP-WCMC & IUCN, 2024). Moreover, connectivity is a moving target. Seasonal shifts in resources can alter movement pathways within a single year, while climate change and human footprint expansion threaten landscape permeability (Brodie et al., 2025; Liczner et al., 2024). Despite a growing recognition of these dynamics, many applications still treat corridor priorities as static, failing to account for anthropogenic and climate impacts (Frans & Liu, 2024; Liczner et al., 2024). This risks a mismatch between conservation strategies and the conditions species will actually face.

Functional connectivity, or the degree to which a landscape facilitates organism movement and maintains ecological processes (Taylor et al., 1993, 2006), can shift markedly over time as physical habitat configuration and species' resource requirements change. This connectivity is essential for facilitating species movement and persistence in the face of global environmental change (Groves et al., 2012; Heller & Zavaleta, 2009; Naidoo et al., 2025). Connected patches of habitat generally support more species and see a reduced rate of local extinctions compared to isolated habitats (Damschen et al., 2019), making functional connectivity central to conservation efforts. Protected areas, although essential for safeguarding habitat, cannot on their own sustain connectivity. Without functional linkages between them, protected populations remain genetically and behaviorally isolated, and climate and land-use change can reduce the suitability of reserves over time (Asamoah et al., 2021; Hoffmann et al., 2019; Naidoo et al., 2025). Wildlife management and conservation approaches beyond PAs have emphasized

connectivity between core habitat through a variety of land designations that allow for continued human and wildlife use, especially across the broader landscape matrix surrounding strictly protected areas (Giliba et al., 2023; Huang et al., 2024).

Habitat connectivity and the dynamic movement strategies of many large mammals are threatened by both anthropogenic climate change and land use conversion (Costanza & Terando, 2019; Mantyka-Pringle et al., 2015). Anthropogenic land use change and resulting habitat fragmentation reduce patch size and increase inter-patch distance, constraining dispersal, while climate change drives range shifts and phenological mismatches that compound the impacts of fragmentation (Haddad et al., 2015; Oliver & Morecroft, 2014; Opdam & Wascher, 2004). This combination limits range shifts as fragmented landscapes prevent populations from tracking suitable climatic conditions following droughts, heat extremes, and shifts in seasonal resource availability (Opdam & Wascher, 2004). Climate change further amplifies the effects of fragmentation because isolated populations are more sensitive to extreme and variable climate (Oliver et al., 2013; van Aarde et al., 2021). At the metapopulation level, the resulting increase in local extinction risks and reduction in recolonization capacity from changes in climate and land cover leads to declines in regional population stability, persistence, and connectivity (Bonnot et al., 2017; Mestre, 2018).

Functional connectivity for savanna elephants is inherently seasonal. Movement is primarily driven by access to green vegetation and surface water, but the relative importance of these resources and the resulting movement patterns shift markedly between wet and dry seasons (Chamaillé-Jammes et al., 2007; Kaszta et al., 2021). One study found a 62% decline in surface water availability across the year in a key elephant conservation landscape in Southern Africa (Schaffer-Smith et al., 2022). This extreme variation in water and forage availability prompts long-distance migrations between dry-season refugia near permanent water sources and wet-season foraging grounds (Chamaillé-Jammes et al., 2007; Wato et al., 2018). This seasonal

migration is critical for maintaining consistent access to food and water, and thus represents a fundamental process linking populations across protected areas. Future climate projections indicate shorter rainy seasons, prolonged drought, and rising evapotranspiration rates that are expected to substantially reduce ephemeral surface water persistence through the mid-century (Nooni et al., 2022; Steinberg, 2023; Yahaya et al., 2024), with direct implications for the seasonal resistance surfaces used in our analysis. Consequently, seasonal variation in connectivity is a fundamental feature of movement ecology of migratory megafauna, with direct consequences for population viability (Avagar et al., 2014; Harris et al., 2009). Accounting for seasonality in current and future connectivity is therefore of critical importance for identifying realized impacts to African savannah elephants.

Connectivity modeling can identify species' movement patterns to guide strategic conservation planning across changing landscapes (Correa Ayram et al., 2016; B. H. McRae et al., 2008), but accurate predictions require knowledge of present-day connectivity, including seasonal movement patterns, as well as projected future changes. While some studies have empirically modeled seasonal migration dynamics (McClure et al., 2016; Poor et al., 2012), reviews have shown that many others remain largely theoretical, lacking empirical validation (Creech et al., 2024; Laliberté & St-Laurent, 2020). Additionally, a review has noted that few studies fully integrate the combined effects of future climate and land-use change (Costanza & Terando, 2019), which are particularly critical in regions where anthropogenic pressures are expanding. Here we present a seasonally-explicit connectivity forecasting workflow that integrates coupled climate-development scenarios (Shared Socioeconomic Pathways - Representative Concentration Pathways; SSP-RCP) with seasonal resource dynamics to project how functional connectivity may shift over coming decades.

We develop and demonstrate such a workflow using African savannah elephants (*Loxodonta africana*) in the Kavango-Zambezi Transfrontier Conservation Area (KAZA) in Southern Africa.

KAZA was established with the explicit goal of conserving biodiversity through landscape connectivity, particularly for wide-ranging megafauna (KAZA Elephant Sub Working Group, 2023; Stoldt et al., 2020), making it an ideal system for testing connectivity forecasting methods. Elephants can serve as a focal species for conservation planning (Brennan et al., 2020): their extensive spatial needs and co-occurrence with other large mammals, including ungulates and carnivores, make them a useful proxy for identifying corridors that can benefit multiple species (Epps et al., 2011). At the same time, KAZA faces mounting pressures from settlements, agricultural activities, roads, and veterinary fences, which already threaten habitat connectivity (Brennan et al., 2020; Naidoo et al., 2022, 2024). Anthropogenic influences are only expected to grow: sub-saharan Africa is projected to account for the majority of the increase in the global population over the rest of this century, adding about 23–38 million more people to the population per year (Gu et al., 2021). Thus, KAZA is an ideal focal system for modeling the combined impacts of anthropogenic climate and land use change on a well-understood migratory species.

We address three research questions:

- i) How much, where, and in what directions will land cover change in KAZA under a range of future coupled climate-development scenarios?
- ii) How will projected land-cover change affect connectivity between large, stable elephant populations in protected areas?
- iii) How does landscape connectivity differ between the wet and dry seasons, and how is seasonal connectivity projected to shift under future climate and land use change?

To answer these questions, we leveraged data on projected landscape characteristics to analyse connectivity, accounting for seasonal shifts and uncertainty in long-term trends. We modeled elephant movement and habitat connectivity under future paired SSP-RCP scenarios,

evaluated model parameter assumptions through a sensitivity analysis, and validated model outputs using GPS-collared elephant locations. Our approach can be applied to model future connectivity shifts for a wide range of species and landscapes beyond KAZA.

2. Materials and methods

2.1. Study System

The Kavango-Zambezi Transfrontier Conservation Area (KAZA) (Fig. 1) spans approximately 520,000 km² across Zambia, Zimbabwe, Botswana, Namibia, and Angola (KAZA Elephant Sub Working Group, 2023). Its mosaic of national parks, game reserves and community conservancies supports the world's largest African savannah elephant metapopulation (~230,000 individuals) (Bussière & Potgieter, 2023), as well as populations of other globally significant megafauna such as the African wild dog, roan and sable antelopes, and African buffalo (Stoldt et al., 2020). KAZA's semi-arid climate is a key determinant in elephant movement (Makati et al., 2025; Polansky et al., 2015), leading to seasonally-dependent movement patterns in the region.

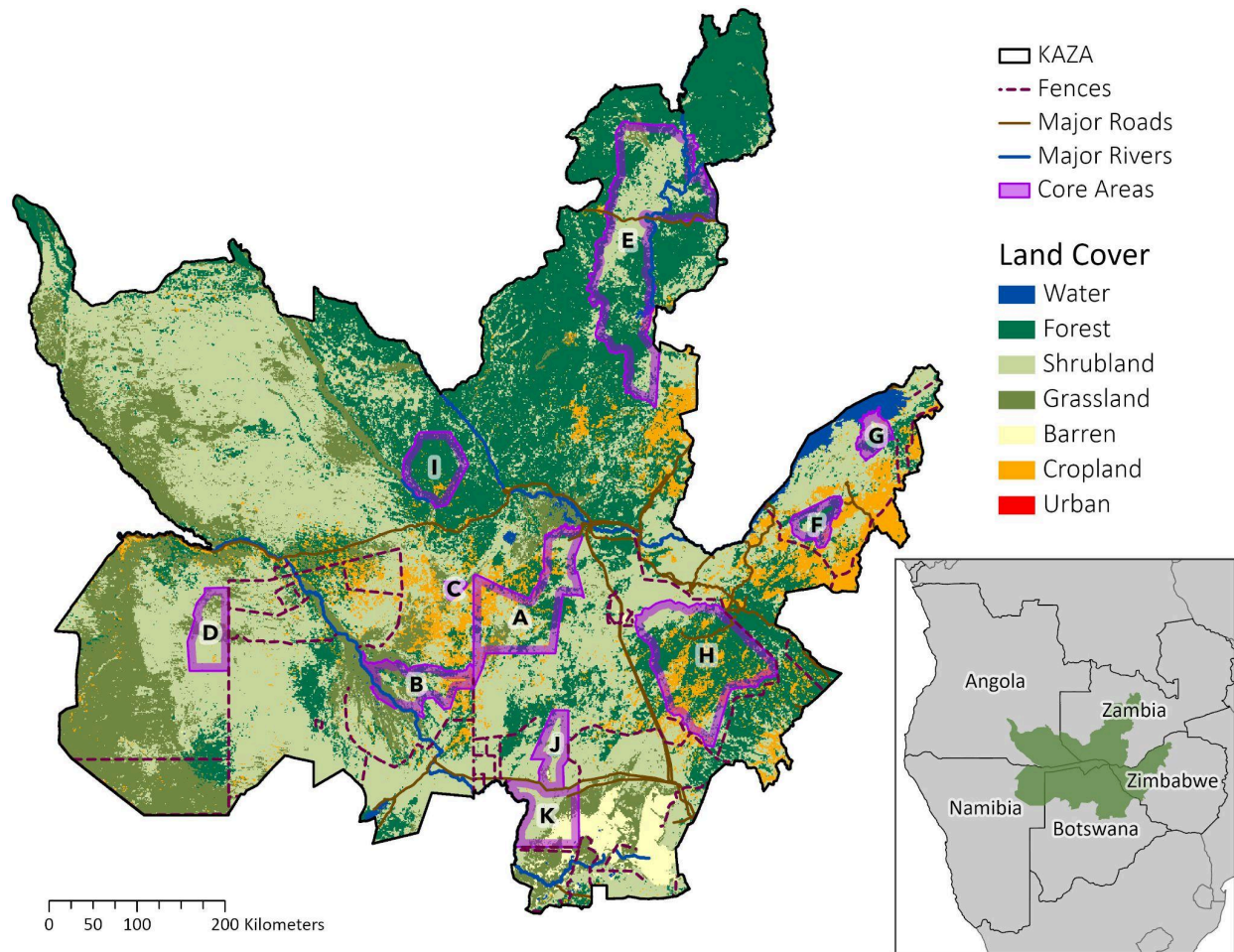


Figure 1. Study Area. Map of the KAZA region with 2015 land cover classifications, veterinary fences, major roads, major rivers, and select protected core habitat areas labeled A through K as follows: A) Chobe B) Moremi C) Nkasa Rupara D) Khaudum E) Kafue F) Chizarira G) Matusadona H) Hwange I) Sioma Ngwezi J) Nxai Pan and K) Makgadikgadi Pans.

2.2. Land Cover Change

To evaluate present and future elephant habitat connectivity, we assessed projected land cover change in KAZA from 2015 to 2050. Three Intergovernmental Panel on Climate Change (IPCC) SSP-RCP landcover projections were selected: SSP1-RCP2.6 (sustainable international development with low radiative forcing), SSP2-RCP4.5 (middle of the road development with

moderate radiative forcing), and SSP3-RCP7.0 (regional rivalry with high radiative forcing) (Calvin et al., 2023). SSP1-RCP2.6 and SSP3-RCP7.0 were chosen to represent the upper and lower ranges of potential future outcomes. In alignment with the nearest land cover dataset time-step, 2015 was selected as the baseline year for this analysis, and changes through 2050 were evaluated for all SSP-RCP scenarios. While our results focus on the SSP1-RCP2.6 and SSP3-RCP7.0 scenarios and the 2050 time horizon, data for 2035 and for SSP2-RCP4.5 are reported in the Supplementary Information.

Land cover data in KAZA was sourced from (Chen et al., 2022), who presented the 2015 land cover baseline and modeled global land cover change across various SSP-RCP scenarios and provided land cover data containing 20 different plant functional types (PFTs) at 1 km resolution. These PFTs were simplified into the seven land cover categories indicated in Fig. 1 to align with the resistance layer methodology (Van de Perre et al., 2014), discussed further in section 2.3.

2.3. Least-Cost Path Connectivity Analysis

Habitat connectivity was assessed with least-cost path (LCP) analysis, which uses cost-weighted distance (CWD) to estimate the least energetically expensive paths individuals may take between habitat patches (Savary et al., 2022). Cost-distance analysis expresses travel effort in distance-equivalence units, allowing for the identification of corridors where movement is impeded. LCPs represent the most efficient route between identified habitat patches given the resistance of the landscape. LCP and CWD outputs can be used to predict how elephants are likely to move across a landscape and design conservation interventions to support that movement. We used Linkage Mapper v3.1.0 (toolbox for ArcGIS 3.1.0) (B. McRae & Kavanagh, 2011) to conduct our LCP and CWD analysis.

Two inputs are required: core area polygons, representing the suitable areas of protected habitat supporting stable populations between which elephants can move (Cushman et al., 2010), and a raster representing the relative resistance each landscape feature poses to elephant movement.

Core Areas

Core areas of elephant habitat were defined based on (Huang et al., 2024), who identified 11 priority areas across KAZA. Their selection was based upon their protection status per the International Union for the Conservation of Nature (IUCN) World Database of Protected Areas (WDPA), published in April 2024, and permanent presence of elephant populations (Supplemental S.3). The core areas are Chobe National Park, Moremi Game Reserve, Nkasa Rupara National Park, Khaudum National Park, Kafue National Park, Chizarira National Park, Matusadona National Park, Hwange National Park, Sioma Ngwezi National Park, Nxai Pan National Park, and Makgadikgadi Pans National Park (Fig. 1). The Core Areas were assumed to remain stable throughout the study period.

Resistance Layers

We developed resistance layers that integrate multiple spatial data layers, representing land cover classifications and landscape features known to influence elephant movement. Each land cover type and landscape feature is assigned a value that represents the resistance that it poses to movement. Using these values, movement is modeled between two or more areas identified as high-quality habitat or home range (Savary et al., 2022). Using the *terra* package (version 1.8-60) (Hijmans et al., 2025) in R (version 4.4.1) (R Core Team, 2024), 18 separate resistance rasters were created to encompass three time-steps (2015, 2035, and 2050) and two seasons, with 2015 representing baseline conditions and 2035 and 2050 reflecting projections

under three SSP-RCP scenarios. Our approach builds on the methods established by (Van de Perre et al., 2014) who developed wet and dry season resistance layers for elephants in Tanzania based on previous literature that indicated a strong seasonality in movement patterns. Landscape features used to build the resistance layer were fences, roads, slope, rivers, ephemeral waterholes (EWHs), and designated core areas. Veterinary fences, erected to separate wildlife from livestock to limit disease transmission, pose a formidable barrier to elephant movement (Naidoo et al., 2022). We sourced veterinary fence locations from Huang et al. (2022) and Taylor (2019). Roads often can have adverse effects on elephant movement (Blake et al., 2008). Road shapefiles were obtained from the Global Roads Inventory Project (GRIP). Slope was included because elephants generally avoid steep terrain (Epps et al., 2013), and was derived from the USGS 3-arc-second DEM for Africa (Verdin, 2017). Because the KAZA landscape encompasses substantial semi-arid regions alongside wetter floodplains, the characterization of perennial and ephemeral surface water is essential in predicting wildlife movement (Naidoo et al., 2020; Schaffer-Smith et al., 2022). Large, fast-flowing rivers can act both as attractants and barriers to movement while smaller rivers primarily attract and are easily crossed by elephants (Naidoo et al., 2022). River shapefiles were sourced from the HydroRIVERS v1.0 dataset (Lehner & Grill, 2013), with permanence classified from flow order following Huang et al. (2022). All of those features were assumed to remain stable throughout the study period.

EWHs, small surface water features that appear seasonally throughout the landscape, are strong predictors of elephant path selection (Naidoo et al., 2020). Similarly, perennial waterholes can influence elephant movement patterns (Polansky et al. 2015). EWH frequency layers were generated in Google Earth Engine using the workflow of Schaffer-Smith et al., 2022. To account for future aridification, EWHs were projected out to 2050 under the different climate scenarios

(Supplemental S1.2). All datasets were rasterized and resampled to the same coordinate reference system (EPSG 3857) at 250 m resolution within the KAZA boundary.

We first calculated cumulative resistance ranks – relative levels of elephant movement difficulty – for each pixel across the landscape (Table 1). Landscape features that increase resistance (e.g., steep slopes, roads, fences, river barriers) increased the rank, while features that facilitate movement and attract elephants (e.g., seasonal rivers, EWH, core areas) decreased the rank. Resistance ranks from (Van de Perre et al., 2014) were assigned to land cover classifications and landscape features in our dataset (Table S1). Several resistance ranks were modified based on current literature to more accurately represent landscape conditions in KAZA (Table 1).

After calculating the final resistance rank per pixel, ranks were converted to resistance values for use in connectivity modeling (Table S2). These values were adapted from Van de Perre et al. (2014) and ranged from 0.1 (easiest for elephant movement) to 2000 (most difficult for elephant movement). A detailed description of resistance rank calculations is provided in the Supplementary (S1).

For features where the literature did not have a consensus, we performed a sensitivity analysis to quantify the influence of individual features and rank assignments on model outputs. The findings informed our parameter choices for the baseline model, including the addition of smaller rivers (flow order 4) as barriers to movement, core areas as areas of decreased resistance, and land cover resistance ranks aligned with Van de Perre et al., 2014 (Supplementary S2).

Table 1. Resistance ranks. Superscripts indicate that the landscape feature and/or rank differs from (Van de Perre et al., 2014), and the rationale for the modification is given in footnotes.

Landscape Features	Wet		Dry	
	Rank	Description	Rank	Description
Land Cover Class				
Water	5	Strong impeding effect on movement	5 ^a	Strong impeding effect on movement
Broadleaf deciduous tree, tropical	1	Prime movement habitat	1	Prime movement habitat
Broadleaf deciduous tree, temperate	1	Prime movement habitat	1	Prime movement habitat
Broadleaf deciduous shrub, temperate	2	Secondary habitat for movement	2	Secondary habitat for movement
Mixed C3/C4 grass	1	Prime movement habitat	2	Secondary habitat for movement
Barren	4	Impeding effect on movement	4	Impeding effect on movement
Cropland	3	Limited negative influence on movement	3	Limited negative influence on movement
Urban	5	Strong impeding effect on movement	5	Strong impeding effect on movement
Roads				
Main roads (primary roads)	5	Strong impeding effect on movement	5	Strong impeding effect on movement
Trails (secondary & tertiary roads)	3	Limited negative influence on movement	3	Limited negative influence on movement
Fences				
Permanent veterinary fences	6 ^b	Impermeable barrier	6 ^b	Impermeable barrier
Rivers				
River Barriers (Flow Order ≤ 4)	6 ^c	Impermeable barrier	6 ^c	Impermeable barrier
Small Rivers (Flow Order > 4)	-1 ^d	Significant attractant	-1	Significant attractant
River Buffers				
Permanent river buffer	-1	Significant attractant	-2	Very strong attractant
Slope				
15° – 30°	+1	Prime movement habitat	+1	Prime movement habitat
$> 30^\circ$	6	Impermeable barrier	6	Impermeable barrier
Ephemeral Watering Holes (EWH)				
20 – 40 % frequency	-1	Significant attractant	NA	NA
40 – 80 % frequency	-1	Significant attractant	-1	Significant attractant
80 – 100 % frequency	-1	Significant attractant	-2	Very strong attractant
Core Areas				
Protected areas	-1	Significant attractant	-1	Significant attractant

^a Van de Perre had no permanent water bodies in their study area. Permanent water bodies were thus assigned the same rank as permanent rivers during the wet season.

^b No fences were included in Van de Perre's analysis. Permanent veterinary fences are treated as impermeable barriers (Huang et al., 2022).

^c Van de Perre gives permanent rivers a rank of 5 in wet season and 4 in dry season. Based on literature indicating some larger permanent rivers in KAZA are impassable to most elephants and some smaller permanent rivers pose no resistance to movement, we treat the Zambezi River (flow order 3), the Okavango and Kafue Rivers (flow order 4) as impermeable barriers year round, and do not treat lower flow order permanent rivers (orders 5–7) as barriers (Huang et al., 2022; van Aarde et al., 2021).

^d Van de Perre gives seasonal rivers (flow order of ≥ 7) a rank of 4 in the wet season. Based on literature indicating that seasonal rivers in KAZA attract elephants and generally do not pose barriers to movement, we treat seasonal rivers as attractants instead of barriers (Huang et al., 2022).

Least-Cost Path Parameters

Using the resistance layers and core areas as inputs, we used Linkage Mapper implemented through ArcPy to generate consistent and comparable Least Cost Paths (LCPs), the lowest resistance route between two core areas, in each output. Each season-scenario run produced connections between each core area to every other core area ($n = 55$ LCPs). The Linkage Mapper outputs included shapefiles of the LCPs between core areas and rasters of the CWDs from the LCPs, representing the additional energetic cost of taking a different route between core areas (B. McRae & Kavanagh, 2011).

Model Validation

We validated the baseline connectivity model by comparing CWD values at known elephant locations within the central and western portions of KAZA, extracted from GPS collar data, to those at randomly selected reference locations. We used GPS telemetry data from 65 collared individuals from 2010 to 2020 collected by University of Namibia in Western KAZA. From within a 95% kernel density estimate (KDE), the locations of 1,000 randomly selected collar pings were compared with 1,000 randomly selected points. CWD values were extracted for both sets of points and compared using Welch's two-sample t-tests to assess whether modeled connectivity was significantly higher in areas used by elephants (SI).

2.4. Future Connectivity Change

We assessed projected changes in connectivity between baseline and future scenarios both at the landscape-scale and at the individual core area scale. To quantify these changes, we compared future scenarios with the 2015 baseline runs using three metrics derived from LCP modeling: spatial LCP overlap, LCP length, and CWD. LCP overlap was calculated by buffering

each path by 250 m and measuring the proportion of area shared between the baseline and future versions of each path (Savary et al., 2022). LCP length and CWD were extracted directly from Linkage Mapper outputs. To assess the effects of future scenarios (each scenario–year combination) on each metric, we fit linear regression models with clustered standard errors at the LCP level using the *estimatr* (version 1.0.6) package (Blair et al., 2025) in R. Models were run separately for wet and dry seasons.

To evaluate potential shifts in the connectivity of core elephant reserves, we compared CWD values for LCPs connecting to each of the 11 core areas. Higher CWD values across a core area's LCP indicates increased isolation (Perzanowski et al., 2020). For each season, we calculated the mean CWD of the 10 LCPs entering or exiting each core under the baseline (2015), SSP1-RCP2.6 (2050) and SSP3-RCP7.0 scenario (2050). We calculated the absolute and percent change in mean CWD for each core area across all LCPs and tested for significant differences between baseline and future CWD values using Welch's t-test.

3.0 Results

3.1. Land Cover Change Projections

The amount of land cover change across KAZA from 2015 to 2050 varies by SSP-RCP scenario. SSP1-RCP2.6, the most optimistic scenario, shows modest gains in forest/shrubland cover (+15,300 km²) alongside modest losses in grassland (–12,400 km²) and cropland (–5,600 km²) cover by 2050. SSP3-RCP7.0 has the most drastic rates of land conversion, with widespread losses of forest/shrubland (–50,600 km²) and grassland (–9,200 km²) driven by conversion of natural land cover types to cropland (+59,200 km²; Fig. 2). Tabular results for all scenarios can be found in Table S9.

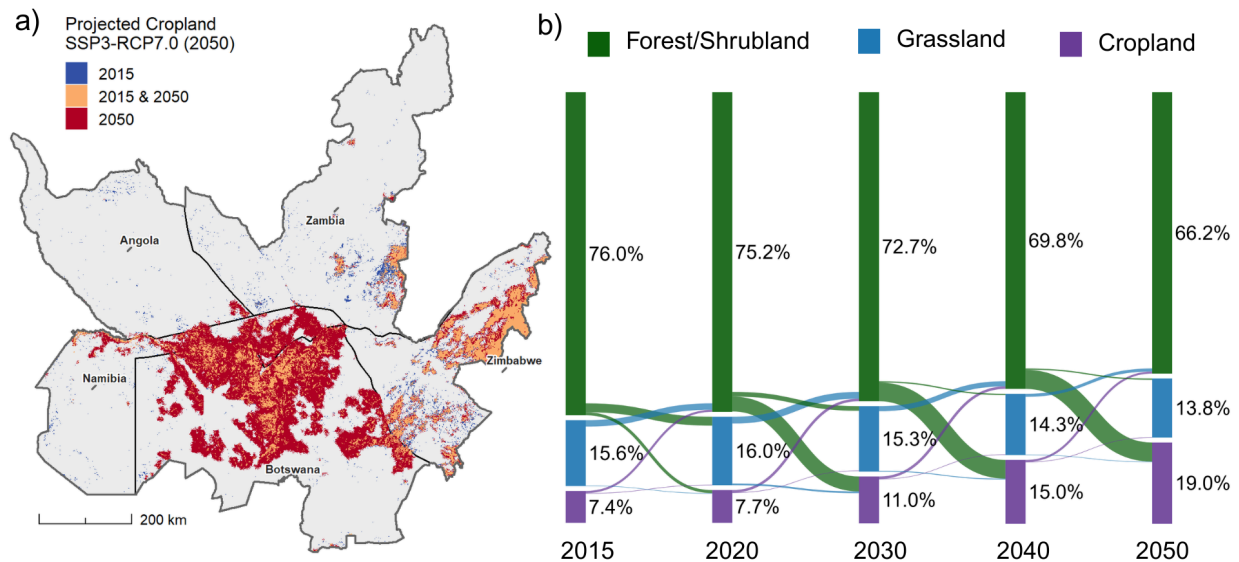


Figure 2. Projected cropland expansion and land-cover change in KAZA from 2015 to 2050 for SSP3-RCP7.0. (a) Spatial distribution of KAZA's cropland in 2015 (yellow & blue), areas projected to be converted to cropland by 2050 (dark red) and abandoned cropland (blue). (b) Sankey diagram summarizing decadal land cover trajectories. Vertical nodes indicate the proportion of KAZA occupied by forest and shrubland (green), grassland (blue), and cropland (purple) at each time step. Curved links depict magnitude and direction of land-cover transitions. Barren, water and urban land cover classes were not included because they make up ~1% of KAZA.

3.2. Baseline Connectivity

The 2015 baseline model outputs (Fig. 3) indicate moderate spatial overlap between wet and dry season LCPs (mean overlap of 28.6% and 26.5%, respectively) across the 55 common linkages. LCPs from Khaudum National Park eastward change significantly during the dry season, traveling further northward. Otherwise, LCPs stay largely consistent on a regional scale, preferring paths within core areas and close to water sources, but diverge at a finer scale.

Central KAZA, especially around and within Chobe National Park, shows several divergent LCPs across both the wet and dry season.

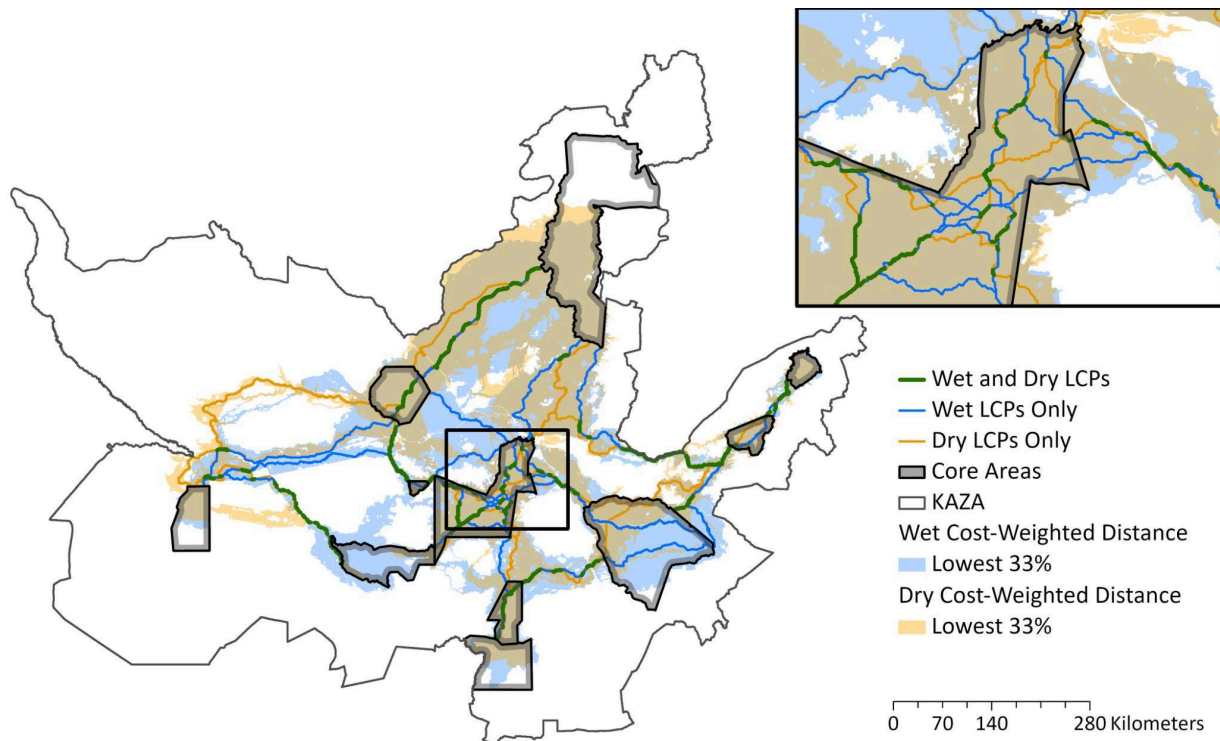


Figure 3. Baseline (2015) landscape connectivity in KAZA. LCPs for the wet season (blue) and dry season (yellow) overlaid on corridors defined by the lowest tercile of CWD values for each season. Green LCP's represent routes generated by both seasonal models. Core areas are outlined in dark gray. The inset enlarges the LCP network around Northern Chobe.

When assessing how well our model predicted actual elephant movement, we found that elephants stayed much closer to the model-generated LCPs than would be predicted by random chance. Elephant collar pings ($n = 65$ elephants, 2010–2020) occurred in pixels with mean CWD approximately 47% lower than random points during the wet season, and 57% lower than random points in the dry season (Table S8). This relationship was significant (Welch $t = 6.87$ and 7.07 , $p < 0.001$; Table S8), suggesting that the model captures landscape features that are predictive of elephant presence.

3.3. Future Connectivity

We assessed potential shifts in elephant habitat connectivity between 2015 and 2050 under SSP1-RCP2.6 and SSP3-RCP7.0 scenarios by predicting changes both in *where* elephants will move (LCP overlaps) and *how difficult* that movement will be (CWD changes). We also identified which core areas are projected to become more isolated in the future.

Landscape-Scale Connectivity Changes

Regression analyses indicated that LCPs in 2050 are projected to significantly deviate spatially from baseline pathways under the SSP3-RCP7.0 scenario in both wet and dry season, particularly during the wet season (Table 2). Under SSP1-RCP2.6, LCPs shift more in the dry season than in the wet season, but overall are more similar to the baseline compared to SSP3-RCP7.0. SSP2-RCP4.5 LCPs indicate intermediate outcomes bracketed by the other scenarios, and are reported in Table S10.

Under SSP3-RCP7.0, mean CWDs increased significantly relative to the baseline in both the wet and dry seasons (Table 2; Fig. 4). Notably, this increase was driven by rising landscape resistance along existing corridors rather than dramatic shifts in corridor geometry; the lengths of LCPs did not significantly change in either season (Table S12). Conversely, mean CWDs for SSP1-RCP2.6 declined in both dry and wet seasons, indicating a slight increase in connectivity between 2015 and 2050.

Table 2. Estimated effects of 2050 scenarios on least-cost-path (LCP) overlap and cost-weighted distance (CWD) relative to the 2015 baseline. We report overlap and CWD coefficients and standard errors from panel regressions with dummy variables for each LCP. See SI Table S10 and S11 for confidence intervals.

Scenario	Season	
	Dry	Wet
LCP Overlap		
SSP1-RCP2.6	-0.27*** (0.02)	-0.44*** (0.03)
SSP3-RCP7.0	-0.43*** (0.04)	-0.81*** (0.03)
CWD		
SSP1-RCP2.6	-22,649*** (2,517)	-17,896*** (3,578)
SSP3-RCP7.0	118,832*** (13,699)	186,814*** (22,460)
Observations: 385		
Significance: *** p < 0.001; ** p < 0.01; * p < 0.05; · p < 0.10.		

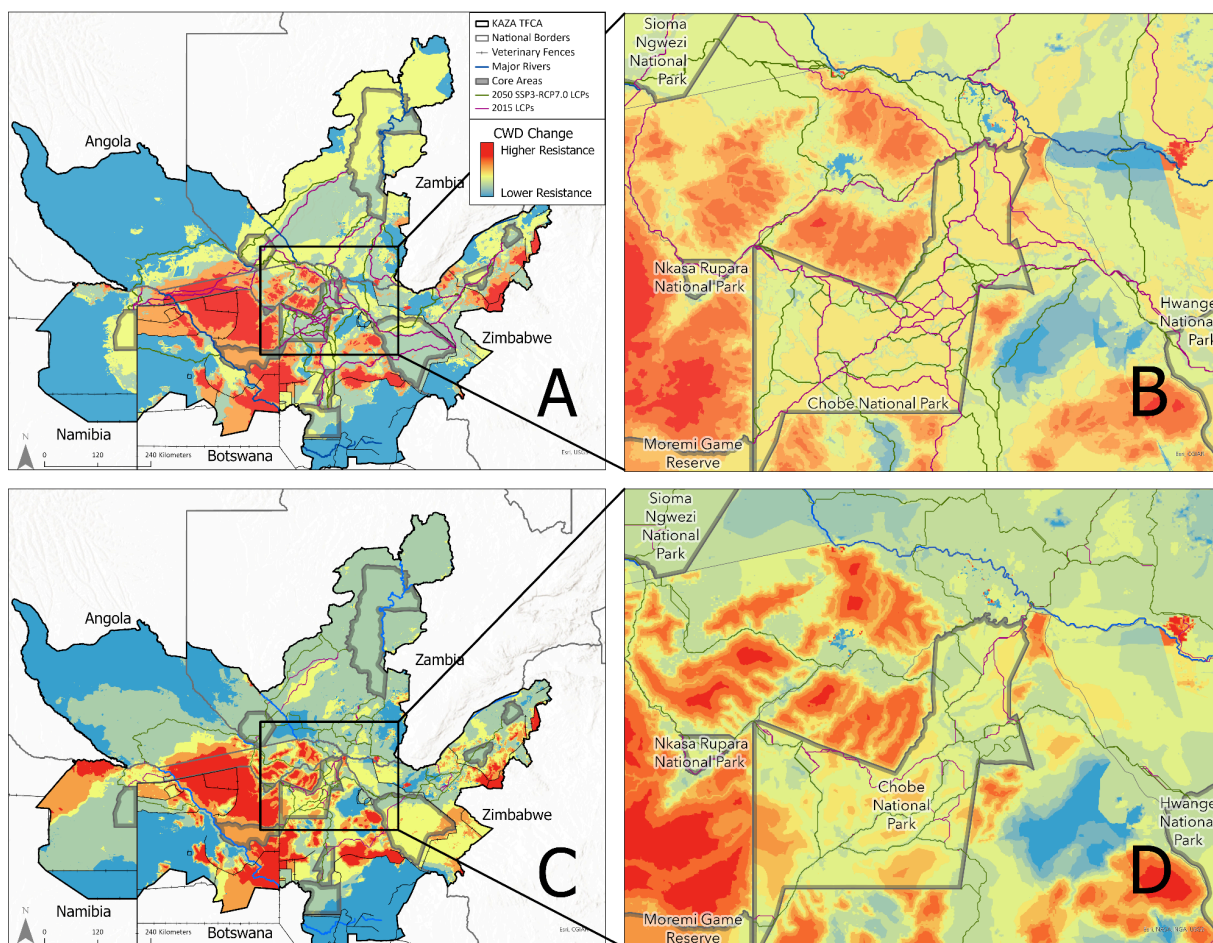


Figure 4. Projected changes in landscape resistance and LCPs between 2015 and 2050 (SSP3-RCP7.0). The panels show (a) CWD change in quintiles across KAZA during the wet season; (b) CWD change in deciles within central KAZA during the wet season; (c) CWD change across KAZA during the dry season; and (d) CWD change within central KAZA during the dry season. Red areas indicate increased movement resistance from 2015 to 2050, while blue areas indicate reduced resistance. Overlaid are 2015 LCPs (dark green lines), 2050 LCPs (magenta), core areas (dark grey), veterinary fences (black), and major rivers (blue). Basemap in panels A and C from ESRI World Hillshade (Esri, n.d.).

Isolation of Individual Core Areas

Under baseline conditions, Khaudum, Chizarira, and Matusadona national parks exhibited the highest mean CWDs across seasons, indicating relatively high baseline isolation (Table 3). These core areas are also geographically distant from other core areas within the landscape.

Under SSP3-RCP7.0, almost all core areas are projected to become significantly more isolated by 2050 (Table 3). In the wet season, all but Chobe and Kafue experienced significant increases in mean CWD ($p < 0.05$). Similarly, all but Chobe and Kafue showed significant increases of CWD in the dry season (Table S14).

Several core areas exhibited especially large increases in isolation under SSP3-RCP7.0. Moremi, Nkasa Rupara, Hwange, Sioma Ngwezi, Nxai Pan, and Makgadikgadi Pans each experienced mean CWD increases of 15% or more across both seasons. Khaudum showed a 24.5% increase in the wet season but less than 10% in the dry season. In contrast, central and northeastern core areas showed comparatively small changes in isolation. Chizarira and Matusadona, which were among the most isolated in the baseline scenario, remained among the most isolated under SSP3-RCP7.0.

Conversely, under SSP1-RCP2.6, none of the core areas are expected to become more isolated. Mean CWD is expected to remain stable or decline slightly, with dry season changes spanning -0.9% to -2.9% and wet-season changes ranging from +0.3% to -2.8%, indicating a slight improvement in connectivity under the more sustainable scenario by 2050 (Table S15).

Table 3. Mean cost-weighted distance per core area in 2015 vs 2050 (SSP3-RCP7.0). The average cost-weighted distance (CWD) of the 10 least-cost paths between each core and all other cores is shown for the 2015 baseline and for the 2050 SSP3-RCP7.0 climate scenario, separated by dry and wet seasons. The percent change gives the relative increase in CWD from 2015 to 2050. Higher percentages indicate greater projected isolation of that core area.

Core Area	Dry Season			Wet Season		
	2015	2050	% Change	2015	2050	% Change
Chobe	653,091	666,795	2.1	640,149	699,007	9.2
Moremi	687,350	808,954	17.7	696,892	869,949	24.8
Nkasa Rupara	713,299	827,990	16.1	710,418	932,945	31.3
Khaudum	1,736,953	1,863,822	7.3	1,593,875	1,984,410	24.5
Kafue	1,136,182	1,200,235	5.6	1,190,639	1,275,492	7.1
Chizarira	1,653,569	1,777,496	7.5	1,678,638	1,822,706	8.6
Matusadona	1,777,617	1,905,335	7.2	1,850,370	2,013,728	8.8
Hwange	715,312	859,233	20.1	689,214	840,459	21.9
Sioma Ngwezi	806,264	932,013	15.6	827,328	1,037,860	25.4
Nxai Pan	719,396	894,396	24.3	701,699	927,150	32.1
Makgadikgad Pans	780,631	950,544	21.8	766,005	996,478	30.1

4.0 Discussion

Global change is likely to strain connectivity across protected-area networks without active corridor conservation action.

Using our connectivity forecasting workflow, we show that anthropogenic land use conversion and climate-driven reductions in surface water could dramatically increase the energetic cost of elephant movement among KAZA's core protected areas by 2050. Impacts were most

pronounced during the wet season and under the most severe climate scenario chosen (SSP3-RCP7.0), whereas connectivity remained relatively stable under SSP1-RCP2.6. This divergence highlights that future impacts are scenario-contingent rather than inevitable, and that corridor management that takes into account multiple scenarios can be a critical part of securing long-term connectivity in transfrontier landscapes.

Under SSP3-RCP7.0, 11.6% (59,000 km²) of KAZA's natural land cover is projected to be converted to cropland by 2050. The advance of cropland into corridor-adjacent lands tends to constrict movement routes, raise conflict risk, and sever functional connectivity between protected areas (Brennan et al., 2022; Naidoo et al., 2024). Several core areas in KAZA, particularly Moremi, Hwange, Nxai Pan, and Makgadikgadi Pans, are projected to become significantly more isolated under this scenario, lowering the probability of elephant immigration and emigration. As these PAs are located along the southern portion of KAZA, connectivity decreases may hinder not only elephant movement between reserves, but also elephant movement between KAZA and more southern reaches of Africa. Reduced connectivity among these populations could slow gene flow and restrict the behavioural flexibility under a warmer, drier climate (Lohay et al., 2020).

Our models highlight the importance of seasonality in both baseline and future connectivity. Baseline corridor locations differed strongly between seasons (mean wet-dry LCP overlap 26–29%), consistent with work showing elephant movement is shaped by seasonal water and forage dynamics (Chamaillé-Jammes et al., 2007; Kaszta et al., 2021). Our validation supports this pattern: collar-derived elephant locations fell in areas of significantly lower CWD than random points in both seasons. Critically, future projections reveal asymmetric connectivity loss between seasons. Under SSP3-RCP7.0, wet-season LCPs deviate more strongly from baseline paths and show larger CWD increases than dry-season LCPs. This is a consequence of differing constraints on movement. During the dry season, elephants are anchored to permanent

water sources and major river systems, producing corridors that are spatially constrained regardless of surrounding land cover change, whereas wet-season movement is more dispersed and therefore more vulnerable to cropland expansion and reduced ephemeral water availability.

In KAZA, wet-season migrations from Hwange extend beyond protected boundaries into Botswana, creating inter-PA linkages that exist primarily during the rains (Tshipa et al., 2017). This implies that movement between core areas primarily occurs during the wet season, and that disruptions to wet-season corridors may disproportionately affect metapopulation cohesion. As aridification progresses, wet-season movement could increasingly resemble dry-season patterns, compressing the window of realized landscape connectivity for elephants.

Ephemeral water holes, which act as elephant attractants and increase the permeability of the landscape to elephant movement, play a central role in elephant connectivity that may be threatened by increasing aridity. As natural surface water becomes less persistent, elephants may concentrate movement near remaining water points, potentially increasing conflict with rural households dependent on permanent water sources (Shaffer et al., 2019). Taken together, the cross-season and cross-scenario persistence of some corridor segments provides a ‘no-regrets’ signal for near-term protection under climate-development uncertainty (Schloss et al., 2022), while corridors that shift strongly between seasons or scenarios may require more adaptive management.

Several sources of uncertainty may influence model precision. First, parameter and baseline data limitations, including resistance values adapted from studies in Tanzania and a 2015 baseline approximately a decade behind the present-day, may not fully reflect elephant responses or current landscape conditions in KAZA. Second, spatial and thematic resolution constraints, such as resampling all layers to 250 m and using river data without seasonality, may

obscure fine-scale features and hydrologic variability. Third, static representation of dynamic features, including roads, fences, rivers, and surface-water bodies, limits the model's ability to capture seasonal and long-term climate–development dynamics. Incorporating updated datasets, planned infrastructure changes, and seasonally explicit hydrologic models would improve both accuracy and management relevance.

Despite these limitations, this workflow provides a framework for evaluating intervention scenarios directly. For example, incorporating planned fence removals into future resistance surfaces would allow managers to quantify the impact of removing a particular fence section on connectivity across variable climate-development scenarios. Similarly, the placement and seasonal operation of artificial water points (AWPs) could be incorporated into the workflow to understand how supplemental water provision affects corridors. This is particularly relevant given the ongoing debate around AWP in southern Africa: while seasonal pumping that mimics natural ephemeral pans may incentivize migration and reduce human–elephant conflict (Perkins, 2019; Smith et al., 2024), year-round provision risks anchoring populations within protected areas and disrupting inter-PA connectivity (Perkins, 2020). Comparisons of intervention performance across scenarios can help identify corridor investments that remain robust to multiple climate-development trajectories, complementing thoughtful land-use planning and nature-based livelihood alternatives that reduce human footprint expansion in corridor-adjacent lands (Giliba et al., 2023; Tiller et al., 2025).

5.0 Conclusions

We present a workflow for projecting seasonal functional connectivity under coupled climate and socioeconomic scenarios, and demonstrate its application to African savanna elephants in KAZA. The most severe climate-development scenario (SSP3-RCP7.0) saw substantial

increases in the energetic cost of movement between protected areas by 2050, with the strongest impacts during the wet season, the period when elephants are most likely to migrate long distances. In contrast, connectivity remains stable or improves under SSP1-RCP2.6, underscoring that projected declines are contingent on the realized future climate-development trajectory. For KAZA, our findings identify priority targets for proactive connectivity planning: the core areas facing the greatest projected isolation, corridor bottlenecks where resistance increases without major rerouting, and seasonal water availability as a key driver of wet-season connectivity. Our methodology can be extended to evaluate specific interventions such as AWP placement or planned fence removal by adding them to future resistance surfaces and comparing their effects across scenarios. This approach has broad applicability across migratory species and landscapes where seasonal resource tracking drives movement, providing critical insights into threats and opportunities for habitat connectivity under escalating global change pressures.

Author Contributions

Olivia Hemond, Dana Cohen-Kaplan and Vikesh Dheeriya conceived the ideas, designed methodology, and analysed the data; Olivia Hemond and Evan Patrick led the writing of the manuscript. Evan Patrick, Nickolas McManus, Joana Krieger, Ashley Larsen, and Patrick Rheordanz provided consistent mentorship and feedback throughout the research process. Ezequiel Fabiano and Colgar Sikopo provided insights into the local context, guided methodological choices, and supported model validation. All authors contributed critically to the drafts and gave final approval for publication.

Statement on inclusion

Acknowledgements

The authors gratefully acknowledge funding from the John and Jody Arnhold UC Santa Barbara-Conservation International Climate Solutions Collaborative.

Conflict of Interest

The authors declare no conflicts of interest.

Data availability statement

Data and replication code is available at the following repository [insert DOI & cite]

References

- Asamoah, E. F., Beaumont, L. J., & Maina, J. M. (2021). Climate and land-use changes reduce the benefits of terrestrial protected areas. *Nature Climate Change*, 11(12), 1105–1110. <https://doi.org/10.1038/s41558-021-01223-2>
- Avgar, T., Street, G., & Fryxell, J. M. (2014). On the adaptive benefits of mammal migration. *Canadian Journal of Zoology*, 92(6), 481–490. <https://doi.org/10.1139/cjz-2013-0076>
- Blair, G., Cooper, J., Coppock, A., Humphreys, M., Sonnet, L., & Fultz, N. (2025). *estimatr: Fast Estimators for Design-Based Inference* (Version Version 1.0.6) [Computer software]. <https://cran.r-project.org/web/packages/estimatr/index.html>
- Blake, S., Deem, S. L., Strindberg, S., Maisels, F., Momont, L., Isia, I.-B., Douglas-Hamilton, I., Karesh, W. B., & Kock, M. D. (2008). Roadless Wilderness Area Determines Forest Elephant Movements in the Congo Basin. *PLOS ONE*, 3(10), e3546. <https://doi.org/10.1371/journal.pone.0003546>
- Bonnot, T. W., Thompson III, F. R., & Millspaugh, J. J. (2017). Dynamic-landscape metapopulation models predict complex response of wildlife populations to climate and landscape change. *Ecosphere*, 8(7), e01890. <https://doi.org/10.1002/ecs2.1890>
- Brennan, A., Beytell, P., Aschenborn, O., Du Preez, P., Funston, P. J., Hanssen, L., Kilian, J. W., Stuart-Hill, G., Taylor, R. D., & Naidoo, R. (2020). Characterizing multispecies

- connectivity across a transfrontier conservation landscape. *Journal of Applied Ecology*, 57(9), 1700–1710. <https://doi.org/10.1111/1365-2664.13716>
- Brennan, A., Naidoo, R., Greenstreet, L., Mehrabi, Z., Ramankutty, N., & Kremen, C. (2022). Functional connectivity of the world's protected areas. *Science*, 376(6597), 1101–1104. <https://doi.org/10.1126/science.abl8974>
- Brodie, J. F., Gonzalez, A., Mohd-Azlan, J., Nelson, C. R., Tabor, G., Vasudev, D., Zeller, K. A., & Fletcher, R. J. (2025). A well-connected Earth: The science and conservation of organismal movement. *Science*, 388(6745), eadn2225. <https://doi.org/10.1126/science.adn2225>
- Bussière, E., & Potgieter, D. (2023). *KAZA Elephant Survey 2022, Volume I: Results and Technical Report*. KAZA TFCA Secretariat. https://assets.worldwildlife.org/www-prd/documents/5si83x57zx_1._KAZA_Elephant_Survey_Volume_I.pdf
- Calvin, K., Dasgupta, D., Krinner, G., Mukherji, A., Thorne, P. W., Trisos, C., Romero, J., Aldunce, P., Barrett, K., Blanco, G., Cheung, W. W. L., Connors, S., Denton, F., Diongue-Niang, A., Dodman, D., Garschagen, M., Geden, O., Hayward, B., Jones, C., ... Péan, C. (with Lee, H.). (2023). *Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (First). Intergovernmental Panel on Climate Change (IPCC). <https://doi.org/10.59327/IPCC/AR6-9789291691647>
- Chamaillé-Jammes, S., Valeix, M., & Fritz, H. (2007). Managing heterogeneity in elephant distribution: Interactions between elephant population density and surface-water availability. *Journal of Applied Ecology*, 44(3), 625–633. <https://doi.org/10.1111/j.1365-2664.2007.01300.x>

- Chen, G., Li, X., & Liu, X. (2022). Global land projection based on plant functional types with a 1-km resolution under socio-climatic scenarios. *Scientific Data*, 9(1), 125.
<https://doi.org/10.1038/s41597-022-01208-6>
- Correa Ayram, C. A., Mendoza, M. E., Etter, A., & Salicrup, D. R. P. (2016). Habitat connectivity in biodiversity conservation: A review of recent studies and applications. *Progress in Physical Geography: Earth and Environment*, 40(1), 7–37.
<https://doi.org/10.1177/0309133315598713>
- Costanza, J. K., & Terando, A. J. (2019). Landscape Connectivity Planning for Adaptation to Future Climate and Land-Use Change. *Current Landscape Ecology Reports*, 4(1), 1–13.
<https://doi.org/10.1007/s40823-019-0035-2>
- Creech, T. G., Brennan, A., Faselt, J., Stabach, J. A., & Keeley, A. T. H. (2024). Validating Connectivity Models: A Synthesis. *Current Landscape Ecology Reports*, 9(4), 120–134.
<https://doi.org/10.1007/s40823-024-00102-8>
- Cushman, S. A., Chase, M., & Griffin, C. (2010). Mapping Landscape Resistance to Identify Corridors and Barriers for Elephant Movement in Southern Africa. In S. A. Cushman & F. Huettmann (Eds.), *Spatial Complexity, Informatics, and Wildlife Conservation* (pp. 349–367). Springer Japan. https://doi.org/10.1007/978-4-431-87771-4_19
- Damschen, E. I., Brudvig, L. A., Burt, M. A., Fletcher, R. J., Haddad, N. M., Levey, D. J., Orrock, J. L., Resasco, J., & Tewksbury, J. J. (2019). Ongoing accumulation of plant diversity through habitat connectivity in an 18-year experiment. *Science*, 365(6460), 1478–1480.
<https://doi.org/10.1126/science.aax8992>
- Epps, C. W., Mutayoba, B. M., Gwin, L., & Brashares, J. S. (2011). An empirical evaluation of the African elephant as a focal species for connectivity planning in East Africa. *Diversity and Distributions*, 17(4), 603–612. <https://doi.org/10.1111/j.1472-4642.2011.00773.x>

- Epps, C. W., Wasser, S. K., Keim, J. L., Mutayoba, B. M., & Brashares, J. S. (2013). Quantifying past and present connectivity illuminates a rapidly changing landscape for the African elephant. *Molecular Ecology*, 22(6), 1574–1588. <https://doi.org/10.1111/mec.12198>
- Esri. (n.d.). *World Hillshade* [Map]. Esri. Retrieved <https://doc.arcgis.com/en/data-appliance/latest/imagery-elevation/world-hillshade.htm>
- Frans, V. F., & Liu, J. (2024). Gaps and opportunities in modelling human influence on species distributions in the Anthropocene. *Nature Ecology & Evolution*, 8(7), 1365–1377. <https://doi.org/10.1038/s41559-024-02435-3>
- Fryxell, J. M., & Sinclair, A. R. E. (1988). Causes and consequences of migration by large herbivores. *Trends in Ecology & Evolution*, 3(9), 237–241. [https://doi.org/10.1016/0169-5347\(88\)90166-8](https://doi.org/10.1016/0169-5347(88)90166-8)
- Giliba, R. A., Kiffner, C., Fust, P., & Loos, J. (2023). Modelling elephant corridors over two decades reveals opportunities for conserving connectivity across a large protected area network. *PLOS ONE*, 18(10), e0292918. <https://doi.org/10.1371/journal.pone.0292918>
- Groves, C. R., Game, E. T., Anderson, M. G., Cross, M., Enquist, C., Ferdaña, Z., Girvetz, E., Gondor, A., Hall, K. R., Higgins, J., Marshall, R., Popper, K., Schill, S., & Shafer, S. L. (2012). Incorporating climate change into systematic conservation planning. *Biodiversity and Conservation*, 21(7), 1651–1671. <https://doi.org/10.1007/s10531-012-0269-3>
- Gu, D., Andreev, K., & Dupre, M. E. (2021). Major Trends in Population Growth Around the World. *China CDC Weekly*, 3(28), 604–613. <https://doi.org/10.46234/ccdcw2021.160>
- Haddad, N. M., Brudvig, L. A., Clobert, J., Davies, K. F., Gonzalez, A., Holt, R. D., Lovejoy, T. E., Sexton, J. O., Austin, M. P., Collins, C. D., Cook, W. M., Damschen, E. I., Ewers, R. M., Foster, B. L., Jenkins, C. N., King, A. J., Laurance, W. F., Levey, D. J., Margules, C. R., ... Townshend, J. R. (2015). Habitat fragmentation and its lasting impact on Earth's ecosystems. *Science Advances*, 1(2), e1500052. <https://doi.org/10.1126/sciadv.1500052>

- Harris, G., Thirgood, S., Hopcraft, J., Cromsight, J., & Berger, J. (2009). Global decline in aggregated migrations of large terrestrial mammals. *Endangered Species Research*, 7, 55–76. <https://doi.org/10.3354/esr00173>
- Heller, N. E., & Zavaleta, E. S. (2009). Biodiversity management in the face of climate change: A review of 22 years of recommendations. *Biological Conservation*, 142(1), 14–32. <https://doi.org/10.1016/j.biocon.2008.10.006>
- Hijmans, R., Brown, A., & Barbosa, M. (2025). *terra: Spatial Data Analysis* (Version R Package version 1.8-60) [Computer software]. <https://cran.r-project.org/web/packages/terra/index.html>
- Hoffmann, S., Irl, S. D. H., & Beierkuhnlein, C. (2019). Predicted climate shifts within terrestrial protected areas worldwide. *Nature Communications*, 10(1), 4787. <https://doi.org/10.1038/s41467-019-12603-w>
- Huang, R. M., Aarde, R. J. van, Pimm, S. L., Chase, M. J., & Leggett, K. (2022). Mapping potential connections between Southern Africa's elephant populations. *PLOS ONE*, 17(10), e0275791. <https://doi.org/10.1371/journal.pone.0275791>
- Huang, R. M., Maré, C., Guldmond, R. A. R., Pimm, S. L., & Van Aarde, R. J. (2024). Protecting and connecting landscapes stabilizes populations of the Endangered savannah elephant. *Science Advances*, 10(1), eadk2896. <https://doi.org/10.1126/sciadv.adk2896>
- Kaszta, Ž., Cushman, S. A., & Slotow, R. (2021). Temporal Non-stationarity of Path-Selection Movement Models and Connectivity: An Example of African Elephants in Kruger National Park. *Frontiers in Ecology and Evolution*, 9. <https://doi.org/10.3389/fevo.2021.553263>
- KAZA Elephant Sub Working Group. (2023). *KAZA Policy Brief on Elephant Movements and Connectivity*. <https://www.worldwildlife.org/publications/kaza-policy-brief-on-elephant-movements-and-connectivity>

- Laliberté, J., & St-Laurent, M.-H. (2020). Validation of functional connectivity modeling: The Achilles' heel of landscape connectivity mapping. *Landscape and Urban Planning*, 202, 103878. <https://doi.org/10.1016/j.landurbplan.2020.103878>
- Lehner, B., & Grill, G. (2013). Global river hydrography and network routing: Baseline data and new approaches to study the world's large river systems. *Hydrological Processes*, 27(15), 2171–2186. <https://doi.org/10.1002/hyp.9740>
- Liczner, A. R., Pither, R., Bennett, J. R., Bowman, J., Hall, K. R., Fletcher Jr, R. J., Ford, A. T., Michalak, J. L., Rayfield, B., Wittische, J., & Pither, J. (2024). Advances and challenges in ecological connectivity science. *Ecology and Evolution*, 14(9), e70231. <https://doi.org/10.1002/ece3.70231>
- Lohay, G. G., Weathers, T. C., Estes, A. B., McGrath, B. C., & Cavener, D. R. (2020). Genetic connectivity and population structure of African savanna elephants (*Loxodonta africana*) in Tanzania. *Ecology and Evolution*, 10(20), 11069–11089. <https://doi.org/10.1002/ece3.6728>
- Makati, A. A., Songhurst, A., Bennitt, E., Masunga, G. S., McCulloch, G., Stronza, A., Matsika, T. A., & Eckardt, F. D. (2025). Seasonal Variation in Home Range Sizes and Daily Distance to Ephemeral Surface Water for African Savannah Elephant (*Loxodonta africana*) in Eastern Okavango Panhandle, Northern Botswana. *Ecology and Evolution*, 15(1), e70758. <https://doi.org/10.1002/ece3.70758>
- Mantyka-Pringle, C. S., Visconti, P., Di Marco, M., Martin, T. G., Rondinini, C., & Rhodes, J. R. (2015). Climate change modifies risk of global biodiversity loss due to land-cover change. *Biological Conservation*, 187, 103–111. <https://doi.org/10.1016/j.biocon.2015.04.016>
- McClure, M. L., Hansen, A. J., & Inman, R. M. (2016). Connecting models to movements: Testing connectivity model predictions against empirical migration and dispersal data. *Landscape Ecology*, 31(7), 1419–1432. <https://doi.org/10.1007/s10980-016-0347-0>

- McRae, B. H., Dickson, B. G., Keitt, T. H., & Shah, V. B. (2008). Using circuit theory to model connectivity in ecology, evolution, and conservation. *Ecology*, 89(10), 2712–2724.
<https://doi.org/10.1890/07-1861.1>
- McRae, B., & Kavanagh, D. (2011). *Linkage Mapper Connectivity Analysis Software* [Computer software]. The Nature Conservancy. <https://linkagemapper.org>
- Mestre, F. (2018). Synergistic effects of climate change and habitat fragmentation on species range shifts and metapopulation persistence. *Frontiers of Biogeography*, 9(4).
<https://doi.org/10.21425/F59435859>
- Naidoo, R., Aylward, C., Elliott, W., Keeley, A., Kinnaird, M., Knight, M., Papp, C.-R., Thapa, K., & Antelo, R. (2025). From science to impact: Conserving ecological connectivity in large conservation landscapes. *Proceedings of the National Academy of Sciences*, 122(31), e2410937122. <https://doi.org/10.1073/pnas.2410937122>
- Naidoo, R., Beytell, P., Brennan, A., Carter, J., Carter, K. D., Chamaillé-Jammes, S., Chilambe, B., Hoare, R., Iiyambo, N., Jooste, D., Karidozo, M., Kilian, J. W., Madhlamoto, D., Madiri, T. H., McCulloch, G., Monks, N., Mudimba, I., Ngwenya, N., Nyambe, N., ... Songhurst, A. (2024). Landscape connectivity for African elephants in the world's largest transfrontier conservation area: A collaborative, multi-scalar assessment. *Journal of Applied Ecology*, n/a(n/a). <https://doi.org/10.1111/1365-2664.14746>
- Naidoo, R., Beytell, P., Brennan, A., Kilian, W., McCulloch, G., Stronza, A., Taylor, R., Tsholofelo, C., & Songhurst, A. (2022). Challenges to Elephant Connectivity From Border Fences in the World's Largest Transfrontier Conservation Area. *Frontiers in Conservation Science*, 3. <https://doi.org/10.3389/fcosc.2022.788133>
- Naidoo, R., Brennan, A., Shapiro, A. C., Beytell, P., Aschenborn, O., Du Preez, P., Kilian, J. W., Stuart-Hill, G., & Taylor, R. D. (2020). Mapping and assessing the impact of small-scale ephemeral water sources on wildlife in an African seasonal savannah. *Ecological Applications*, 30(8), e02203. <https://doi.org/10.1002/eap.2203>

- Nooni, I. K., Hagan, D. F. T., Ullah, W., Lu, J., Li, S., Prempeh, N. A., Gnitou, G. T., & Lim Kam Sian, K. T. C. (2022). Projections of Drought Characteristics Based on the CNRM-CM6 Model over Africa. *Agriculture*, 12(4), 495. <https://doi.org/10.3390/agriculture12040495>
- Oliver, T. H., Brereton, T., & Roy, D. B. (2013). Population resilience to an extreme drought is influenced by habitat area and fragmentation in the local landscape. *Ecography*, 36(5), 579–586. <https://doi.org/10.1111/j.1600-0587.2012.07665.x>
- Oliver, T. H., & Morecroft, M. D. (2014). Interactions between climate change and land use change on biodiversity: Attribution problems, risks, and opportunities. *WIREs Climate Change*, 5(3), 317–335. <https://doi.org/10.1002/wcc.271>
- Opdam, P., & Wascher, D. (2004). Climate change meets habitat fragmentation: Linking landscape and biogeographical scale levels in research and conservation. *Biological Conservation*, 117(3), 285–297. <https://doi.org/10.1016/j.biocon.2003.12.008>
- Perkins, J. S. (2019). ‘Only connect’: Restoring resilience in the Kalahari ecosystem. *Journal of Environmental Management*, 249, 109420. <https://doi.org/10.1016/j.jenvman.2019.109420>
- Perkins, J. S. (2020). Changing the Scale and Nature of Artificial Water Points (AWP) Use and Adapting to Climate Change in the Kalahari of Southern Africa. In S. O. Keitumetse, L. Hens, & D. Norris (Eds.), *Sustainability in Developing Countries: Case Studies from Botswana’s journey towards 2030 Agenda* (pp. 51–89). Springer International Publishing. https://doi.org/10.1007/978-3-030-48351-7_4
- Perzanowski, K., Bleyhl, B., Olech, W., & Kuemmerle, T. (2020). Connectivity or isolation? Identifying reintroduction sites for multiple conservation objectives for wisents in Poland. *Animal Conservation*, 23(2), 212–221. <https://doi.org/10.1111/acv.12530>
- Polansky, L., Kilian, W., & Wittemyer, G. (2015). Elucidating the significance of spatial memory on movement decisions by African savannah elephants using state–space models.

- Proceedings of the Royal Society B: Biological Sciences*, 282(1805), 20143042.
<https://doi.org/10.1098/rspb.2014.3042>
- Poor, E. E., Loucks, C., Jakes, A., & Urban, D. L. (2012). Comparing Habitat Suitability and Connectivity Modeling Methods for Conserving Pronghorn Migrations. *PLOS ONE*, 7(11), e49390. <https://doi.org/10.1371/journal.pone.0049390>
- R Core Team. (2024). *R: A language and environment for statistical computing* (Version Version 4.4.1) [R Foundation for Statistical Computing]. <https://www.R-project.org/>
- Savary, P., Foltête, J. C., & Garnier, S. (2022). *Cost distances and least cost paths respond differently to cost scenario variations: A sensitivity analysis of ecological connectivity modeling*.
- Schaffer-Smith, D., Swift, M., Killea, A., Brennan, A., Naidoo, R., & Swenson, J. J. (2022). Tracking a blue wave of ephemeral water across arid southern Africa. *Environmental Research Letters*, 17(11), 114063. <https://doi.org/10.1088/1748-9326/ac98d9>
- Schloss, C. A., Cameron, D. R., McRae, B. H., Theobald, D. M., & Jones, A. (2022). “No-regrets” pathways for navigating climate change: Planning for connectivity with land use, topography, and climate. *Ecological Applications*, 32(1), e02468. <https://doi.org/10.1002/eap.2468>
- Shaffer, L. J., Khadka, K. K., Van Den Hoek, J., & Naithani, K. J. (2019). Human-Elephant Conflict: A Review of Current Management Strategies and Future Directions. *Frontiers in Ecology and Evolution*, 6. <https://doi.org/10.3389/fevo.2018.00235>
- Smith, E., Brown, L. R., & Barrett, A. S. (2024). Insights into artificial waterhole utilization patterns by elephants and rhinos: Lessons from a South African Nature Reserve. *PLOS ONE*, 19(10), e0312158. <https://doi.org/10.1371/journal.pone.0312158>
- Steinberg, N. (2023). *Livelihood-Focused Climate Risk Assessment*. USAID Resilient Waters Program.

- Stoldt, M., Göttert, T., Mann, C., & Zeller, U. (2020). Transfrontier Conservation Areas and Human-Wildlife Conflict: The Case of the Namibian Component of the Kavango-Zambezi (KAZA) TFCA. *Scientific Reports*, 10(1), 7964.
<https://doi.org/10.1038/s41598-020-64537-9>
- Taylor, P. D., Fahrig, L., Henein, K., & Merriam, G. (1993). Connectivity Is a Vital Element of Landscape Structure. *Oikos*, 68(3), 571–573. <https://doi.org/10.2307/3544927>
- Taylor, P. D., Fahrig, L., & With, K. A. (2006). Landscape connectivity: A return to the basics. In K. R. Crooks & M. Sanjayan (Eds.), *Connectivity Conservation* (pp. 29–43). Cambridge University Press. <https://doi.org/10.1017/CBO9780511754821.003>
- Taylor, R. (2019, June 12). *KAZA's Elephants: Impacts of Fences and Other Physical Infrastructure*. KAZA Animal Health Sub Working Group Meeting.
<https://www.wcs-ahead.org/kaza-ahswg/presentations-2019/10-taylor-elephants-fences-ahswg-2019.pdf>
- Tiller, L., Humle, T., Amin, R., Humphries, A., Seaman, D., Sitati, N., & Smith, R. (2025). Elephant pathway use in a human-dominated landscape. *Wildlife Biology*, 2025(1), e01204. <https://doi.org/10.1002/wlb3.01204>
- Tshipa, A., Valls-Fox, H., Fritz, H., Collins, K., Sebele, L., Mundy, P., & Chamaillé-Jammes, S. (2017). Partial migration links local surface-water management to large-scale elephant conservation in the world's largest transfrontier conservation area. *Biological Conservation*, 215, 46–50. <https://doi.org/10.1016/j.biocon.2017.09.003>
- UNEP-WCMC, & IUCN. (2024). *Protected Planet Report 2024*. UNEP-WCMC and IUCN.
<https://pp-digital-report-document.s3.eu-west-2.amazonaws.com/Protected+Planet+Report+2024.pdf>
- van Aarde, R. J., Pimm, S. L., Guldemon, R., Huang, R., & Maré, C. (2021). The 2020 elephant die-off in Botswana. *PeerJ*, 9, e10686. <https://doi.org/10.7717/peerj.10686>

- Van de Perre, F., Adriaensen, F., Songorwa, A. N., & Leirs, H. (2014). Locating elephant corridors between Saadani National Park and the Wami-Mbiki Wildlife Management Area, Tanzania. *African Journal of Ecology*, 52(4), 448–457.
<https://doi.org/10.1111/aje.12139>
- Verdin, K. L. (2017). *Hydrologic Derivatives for Modeling and Applications (HDMA) Data* [Dataset]. U.S. Geological Survey. <https://doi.org/10.5066/F7S180ZP>
- Wato, Y. A., Prins, H. H. T., Heitkönig, I. M. A., Wahungu, G. M., Ngene, S. M., Njumbi, S., & van Langevelde, F. (2018). Movement Patterns of African Elephants (*Loxodonta africana*) in a Semi-arid Savanna Suggest That They Have Information on the Location of Dispersed Water Sources. *Frontiers in Ecology and Evolution*, 6.
<https://doi.org/10.3389/fevo.2018.00167>
- Yahaya, I., Zhou, J., Jiang, S., Xu, R. H., Huang, J., Su, B., Jing, C., Qiu, F., & Jiang, T. (2024). Depiction of Drought Over Africa in the Light of Changing Climate from CMIP6 Models. *Earth Systems and Environment*, 8(4), 1011–1035.
<https://doi.org/10.1007/s41748-024-00502-0>
- shiagbor, G., & Danquah, E. (2017). Seasonal habitat use by Elephants (*Loxodonta africana*) in the Mole National Park of Ghana. *Ecology and Evolution*, 7(11), 3784–3795. <https://doi.org/10.1002/ece3.2962>